Myanmar rCARI Validation Study Primary Results

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1 Survey Progress

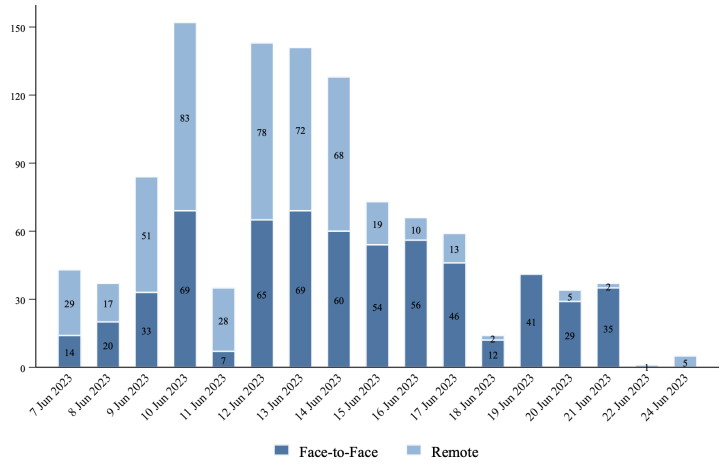
The data for this study was collected from June 7 to June 24, 2023, encompassing 21 camps located in the Kachin and Shan regions. To ensure a representative sample, a pre-collected household list was utilized, from which 1665 households were randomly selected. These selected households were then assigned to two distinct survey modalities: Face-to-Face (F2F) and Remote. Additionally, two different form lengths, namely Short and Long, were randomly assigned to the households. Overall, we completed a total of 1093 completed surveys for the analysis.

Table 1: Sample Completion: Face-to-Face vs. Remote

	Not Completed		Completed		Total	
	Count	Percent	Count	Percent	Count	Percent
Face-to-Face	223	26.77	610	73.23	833	100.00
Remote	349	41.95	483	58.05	832	100.00
Total	572	34.35	1093	65.65	1665	100.00

Table 1 provides an overview of the sample completion rates for the F2F and Remote survey modalities. For F2F survey, a total of 833 households were included in the sample. Out of these, 610 households successfully completed the survey, accounting for 73.23% of the F2F sample. On the other hand, 223 households did not complete the survey, representing 26.77% of the sample. For Remote, 832 households were included in the sample. Of these, 483 households completed the survey, which corresponds to a completion rate of 58.05% for the Remote sample. The remaining 349 households (41.95% of the Remote sample) did not complete the survey due to non-response and availability.

Figure 1: Data Collection Progress by Date and Modality



2 Randomization Quality

To assess the balance of the treatment (Remote) and control (F2F) groups, we examined various respondent and household characteristics, including basic demographics, income sources, and changes over time. This allowed us to evaluate whether the two groups were similar in terms of these factors, ensuring a fair comparison between the treatment and control conditions.

2.1 Modality Randomization

Table 2: Balance Table on Randomized Survey Modality Assignment: Face-to-Face vs. Remote

Variable	(1) F2F Mean/(SE)	(2) Remote Mean/(SE)	(1)-(2) Pairwise t-test Mean difference
Respondent Age	42.24	38.26	3.98***
	(0.54)	(0.62)	
Female Respondent	0.81	0.80	0.01
	(0.02)	(0.02)	
Head of Household Respondent	0.42	0.35	0.07**
	(0.02)	(0.02)	
Household Size	5.52	5.49	0.03
	(0.09)	(0.10)	
Number of household members contributing to income	1.41	1.41	0.00
	(0.03)	(0.04)	
Number of household income sources	1.18	1.24	-0.07*
	(0.03)	(0.03)	
Income Source: None	0.10	0.04	0.06***
	(0.01)	(0.01)	
Income Source: Skilled Waged Labor	0.12	0.08	0.04**
	(0.01)	(0.01)	
Income Source: Unskilled Waged Labor	0.49	0.57	-0.08**
	(0.02)	(0.02)	
Income Source: Trade	0.08	0.04	0.05***
	(0.01)	(0.01)	
Income Source: Agricultural Production	0.12	0.17	-0.05**
	(0.01)	(0.02)	
Income Change: None	0.42	0.41	0.01
	(0.02)	(0.02)	
Income Change: Increase	0.12	0.14	-0.03
	(0.01)	(0.02)	
Income Change: Reduced < 25%	0.23	0.18	0.04*
	(0.02)	(0.02)	
Income Change: Reduced 25% - 50%	0.19	0.18	0.01
	(0.02)	(0.02)	
Income Change: Reduced 50+%	0.05	0.09	-0.03**
	(0.01)	(0.01)	
Number of observations	610	483	1093

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

2.2 F2F Length Randomization

Table 3: Balance Table on Randomized Survey Length Assignment in F2F Survey: Long vs. Short

Variable	N	(1) F2F Mean/(SE)	N	(2) Remote Mean/(SE)	N	(1)-(2) Pairwise t-test Mean difference
Respondent Age	305	41.33	305	43.15	610	-1.81*
		(0.77)		(0.76)		
Female Respondent	305	0.82	305	0.80	610	0.02
		(0.02)		(0.02)		
Head of Household Respondent	305	0.43	305	0.42	610	0.01
		(0.03)		(0.03)		
Single Respondent	305	0.11	305	0.07	610	0.05**
		(0.02)		(0.01)		
Married Respondent	305	0.66	305	0.70	610	-0.05
		(0.03)		(0.03)		
IDP Household	305	0.77	305	0.72	610	0.05
		(0.02)		(0.03)		
Household has disabled member	305	0.17	305	0.20	610	-0.03
		(0.02)		(0.02)		
Number of disabled household members	305	0.18	305	0.23	610	-0.05
		(0.02)		(0.03)		
Head of Household Sex	303	0.60	304	0.68	607	-0.08**
		(0.03)		(0.03)		
Head of Household Age	303	49.05	304	49.88	607	-0.83
		(0.76)		(0.74)		
Household head has attended school	303	0.76	304	0.82	607	-0.06*
		(0.02)		(0.02)		
Household Size	305	5.50	305	5.54	610	-0.04
		(0.13)		(0.13)		
Total number of male household member	305	2.73	305	2.62	610	0.11
		(0.09)		(0.08)		
Total number of female household member	305	2.76	305	2.91	610	-0.15
		(0.08)		(0.09)		
Total number of children	305	2.17	305	2.24	610	-0.07
		(0.08)		(0.09)		
Total number of senior	305	0.46	305	0.48	610	-0.02
		(0.04)		(0.04)		
Total number of adults	305	2.86	305	2.82	610	0.05
		(0.09)		(0.09)		
Household total number of dependents	305	2.63	305	2.72	610	-0.09
		(0.09)		(0.09)		
Household Female/Male Ratio	305	1.69	305	1.63	610	0.06
		(0.11)		(0.10)		
Household Dependency Ratio	305	1.22	305	1.32	610	-0.10
		(0.06)		(0.07)		
Number of household members contributing to income	305	1.49	305	1.34	610	0.15**
		(0.05)		(0.04)		
Number of household income sources	305	1.21	305	1.15	610	0.06
		(0.04)		(0.03)		
Income Source: None	305	0.11	305	0.09	610	0.02
		(0.02)		(0.02)		
Income Source: Skilled Waged Labor	305	0.12	305	0.12	610	-0.01
		(0.02)		(0.02)		
Income Source: Unskilled Waged Labor	305	0.49	305	0.50	610	-0.01
		(0.03)		(0.03)		
Income Source: Trade	305	0.10	305	0.06	610	0.04
		(0.02)		(0.01)		
Income Source: Agricultural Production	305	0.11	305	0.13	610	-0.01
		(0.02)		(0.02)		
Income Change: None	305	0.40	305	0.44	610	-0.03
		(0.03)		(0.03)		
Income Change: Increase	305	0.12	305	0.11	610	0.02
		(0.02)		(0.02)		
Income Change: Reduced < 25%	305	0.25	305	0.20	610	0.05
		(0.02)		(0.02)		
Income Change: Reduced 25% - 50%	305	0.17	305	0.20	610	-0.03
		(0.02)		(0.02)		

2.3 Remote Length Randomization

Table 4: Balance Table on Randomized Survey Length Assignment in Remote Survey: Long vs. Short

Variable	(1) F2F Mean/(SE)	(2) Remote Mean/(SE)	(1)-(2) Pairwise t-test Mean difference
Respondent Age	37.41	39.05	-1.64
	(0.83)	(0.91)	
Female Respondent	0.82	0.78	0.03
	(0.03)	(0.03)	
Head of Household Respondent	0.35	0.35	-0.00
	(0.03)	(0.03)	
Household Size	5.58	5.41	0.17
	(0.15)	(0.13)	
Number of household members contributing to income	1.41	1.41	0.00
	(0.05)	(0.06)	
Number of household income sources	1.26	1.23	0.03
	(0.04)	(0.04)	
Income Source: None	0.03	0.06	-0.03*
	(0.01)	(0.01)	
Income Source: Skilled Waged Labor	0.06	0.09	-0.02
	(0.02)	(0.02)	
Income Source: Unskilled Waged Labor	0.60	0.54	0.06
	(0.03)	(0.03)	
Income Source: Trade	0.03	0.04	-0.00
	(0.01)	(0.01)	
Income Source: Agricultural Production	0.18	0.17	0.01
	(0.03)	(0.02)	
Income Change: None	0.39	0.43	-0.04
	(0.03)	(0.03)	
Income Change: Increase	0.15	0.14	0.01
	(0.02)	(0.02)	
Income Change: Reduced < 25%	0.20	0.17	0.03
	(0.03)	(0.02)	
Income Change: Reduced 25% - 50%	0.20	0.15	0.05
	(0.03)	(0.02)	
Income Change: Reduced 50+%	0.06	0.11	-0.05**
	(0.02)	(0.02)	
Number of observations	233	250	483

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

3 Primary Results

3.1 Impact of Survey Modality on Key Food Security Outcomes

Table 5: Impact of Survey Modality: Face-to-Face vs. Remote

Variable	(1) F2F Mean/(SE)	(2) Remote Mean/(SE)	(1)-(2) Pairwise t-test Mean difference
Food Consumption Score	44.32	43.75	0.57
	(0.50)	(0.55)	
Reduced Consumption Strategies Index (rCSI)	5.43	9.24	-3.81***
	(0.30)	(0.40)	
Maximum Livelihood Coping: None	0.26	0.12	0.14***
	(0.02)	(0.01)	
Maximum Livelihood Coping: Stress level	0.46	0.41	0.05
	(0.02)	(0.02)	
Maximum Livelihood Coping: Crisis level	0.25	0.33	-0.08***
	(0.02)	(0.02)	
Maximum Livelihood Coping: Emergency level	0.03	0.14	-0.10***
	(0.01)	(0.02)	
Total monthly food expenditure (average)	277,712.82	266,487.49	11,225.33
	(5,903.00)	(6,032.45)	
Total monthly non-food expenditure (average)	157,354.79	151,173.19	6,181.61
	(5,956.20)	(4,994.98)	
Monthly total household expenditure	435,067.62	417,660.68	17,406.94
	(9,727.81)	(8,977.45)	
Monthly total household expenditure per capita	85,223.46	84,800.21	423.25
	(1,888.13)	(2,094.78)	
Food Expenditure Share	0.66	0.66	0.01
	(0.01)	(0.01)	
Household being vulnerable using FES CARI	0.47	0.59	-0.12***
	(0.02)	(0.02)	
Household being vulnerable using Income rCARI	0.41	0.54	-0.13***
	(0.02)	(0.02)	
Number of observations	610	483	1093

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table 6: Impact of Survey Length in F2F Survey: Long vs. Short

Variable	(1) Total Mean/(SE)	(2) Long Mean/(SE)	(3) Short Mean/(SE)	(2)-(3) Pairwise t-test Mean difference
Food Consumption Score	44.32	44.67	43.96	0.71
	(0.50)	(0.71)	(0.70)	
Reduced Consumption Strategies Index (rCSI)	5.43	5.70	5.16	0.54
	(0.30)	(0.43)	(0.43)	
Maximum Livelihood Coping: None	0.26	0.25	0.26	-0.01
	(0.02)	(0.02)	(0.03)	
Maximum Livelihood Coping: Stress level	0.46	0.47	0.45	0.02
	(0.02)	(0.03)	(0.03)	
Maximum Livelihood Coping: Crisis level	0.25	0.26	0.24	0.01
	(0.02)	(0.03)	(0.02)	
Maximum Livelihood Coping: Emergency level	0.03	0.02	0.05	-0.02
	(0.01)	(0.01)	(0.01)	
Total monthly food expenditure (average)	277,712.82	300,045.50	255,380.15	44,665.35***
	(5,903.00)	(8,309.95)	(8,202.09)	
Total monthly non-food expenditure (average)	157,354.79	173,898.33	140,811.26	33,087.08***
	(5,956.20)	(9,319.11)	(7,313.85)	
Monthly total household expenditure	435,067.62	473,943.83	396,191.41	77,752.42***
	(9,727.81)	(14,332.85)	(12,797.28)	
Monthly total household expenditure per capita	85,223.46	93,346.30	77,100.63	16,245.67***
	(1,888.13)	(2,940.57)	(2,280.92)	
Food Expenditure Share	0.66	0.66	0.66	-0.00
	(0.01)	(0.01)	(0.01)	
Household being vulnerable using FES CARI	0.47	0.44	0.50	-0.06
	(0.02)	(0.03)	(0.03)	
Household being vulnerable using Income rCARI	0.41	0.41	0.41	-0.01
	(0.02)	(0.03)	(0.03)	
Number of observations	610	305	305	610

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table 7: Impact of Survey Length in Remote Survey: Long vs. Short

Variable	(1) Total Mean/(SE)	(2) Long Mean/(SE)	(3) Short Mean/(SE)	(2)-(3) Pairwise t-test Mean difference
Food Consumption Score	43.75	44.29	43.25	1.04
	(0.55)	(0.79)	(0.77)	
Reduced Consumption Strategies Index (rCSI)	9.24	9.73	8.78	0.95
	(0.40)	(0.63)	(0.51)	
Maximum Livelihood Coping: None	0.12	0.14	0.10	0.04
	(0.01)	(0.02)	(0.02)	
Maximum Livelihood Coping: Stress level	0.41	0.40	0.42	-0.02
	(0.02)	(0.03)	(0.03)	
Maximum Livelihood Coping: Crisis level	0.33	0.34	0.33	0.01
	(0.02)	(0.03)	(0.03)	
Maximum Livelihood Coping: Emergency level	0.14	0.12	0.15	-0.03
	(0.02)	(0.02)	(0.02)	
Total monthly food expenditure (average)	266,487.49	280,289.58	253,623.94	26,665.63**
	(6,032.45)	(8,617.95)	(8,379.99)	
Total monthly non-food expenditure (average)	151,173.19	124,607.94	175,932.00	-51,324.06***
	(4,994.98)	(6,823.42)	(6,911.94)	
Monthly total household expenditure	417,660.68	404,897.52	429,555.94	-24,658.43
	(8,977.45)	(12,570.34)	(12,767.94)	
Monthly total household expenditure per capita	84,800.21	82,083.90	87,331.81	-5,247.91
	(2,094.78)	(2,996.40)	(2,925.90)	
Food Expenditure Share	0.66	0.71	0.61	0.10***
	(0.01)	(0.01)	(0.01)	
Household being vulnerable using FES CARI	0.59	0.63	0.56	0.07*
	(0.02)	(0.03)	(0.03)	
Household being vulnerable using Income rCARI	0.54	0.52	0.56	-0.04
	(0.02)	(0.03)	(0.03)	
Number of observations	483	233	250	483

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.